
Architecture Search and Input Ablations for Single-Subject sEMG Keystroke Decoding

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Abstract

We investigate sEMG-to-keystroke decoding on the emg2qwerty single-subject task, targeting Character Error Rate (CER) minimization while understanding which architectural and data-representation choices drive performance. Starting from a baseline (CER = 28.33%), we evaluate five architectures GRU, BiLSTM, Transformer, Conv+Transformer (Adam), and Conv+Transformer (AdamW + OneCycleLR) alongside ablations on electrode channel subsets, sampling rate, and training data volume. Our best model, a BiLSTM, achieves **13.83%** test CER, a **51.2%** relative improvement. An electrode channel sweep shows monotonically decreasing CER with more channels (25.87% $\rightarrow$ 19.69% $\rightarrow$ 16.51% for 4, 6, 8 channels), with the 4-channel sparse model still outperforming the full 16-channel baseline. We also experimented with sampling rate as well as data volume. Sampling data at 125 Hz had the best test CER, while 62.5 Hz had the best validation CER, indicating that a rate of 125 Hz had the least overfitting. For data volume, we found higher volume of data produced better test and validation CER overall.

1 Introduction

Even within a single subject, sEMG keystroke decoding faces practical barriers the baseline does not address: the performance gap between recurrent and attention-based models in small-data regimes, the cost of a full 16-electrode array, and the minimum data and sampling rate needed for usable performance. We investigate three questions: (1) Do GRU/BiLSTM or Transformer models better capture sEMG temporal dynamics under a fixed, limited training budget? (2) Can a sparse electrode subset match full-array performance? (3) How steeply does CER degrade as training data or sampling rate decreases? Together these address a gap in the emg2qwerty baseline evaluation, which reports only a single architecture on the full dataset at a fixed sampling rate.

2 Methods

2.1 Task and Dataset

All experiments use the emg2qwerty single-subject. Raw sEMG is recorded at 2 kHz across 16 electrodes. A short-time Fourier transform with $n_fft = 64$ yields 33 frequency bins per channel, giving 528-dimensional input features at the full 16-channel configuration. Models are trained on

4-second windows (8000 samples) with [1800, 200] sample padding and evaluated by CER on the held-out test set using CTC loss (2).

2.2 Shared Front-End

All architectures share a **Time-Depth Separable ConvNet (TDS-Conv)** front-end: four blocks with 24 channels and kernel width 32, giving a temporal receptive field of 125 samples (≈ 62.5 ms). The TDS-Conv output is projected through a 384-unit MLP before the sequence encoder. Architectures differ only in their encoder, making comparisons controlled.

2.3 Architectures

Baseline The baseline uses the `TDSConvCTCModule` configuration: 4-layer TDS-Conv front-end with block channels [24, 24, 24, 24], kernel width 32, and MLP features [384], trained with Adam and linear warmup cosine annealing. The baseline achieves 28.33% test CER and serves as the reference point for all comparisons.

GRU Our GRU uses the `TDSConvGruCTCModule` configuration, which extends the identical TDS-Conv front-end with a 4-layer GRU encoder of hidden size 256, trained with Adam and linear warmup cosine annealing. Despite sharing the same convolutional front-end as the baseline, the GRU achieves 18.56% test CER, a 34.5% relative improvement, suggesting that the recurrent encoder provides substantially better sequence modeling than the baseline’s MLP projection alone.

CNN + BiLSTM. CNN encoder followed by a 2-layer Bidirectional LSTM trained with CTC loss (`CNNBiLSTMCTCModule`). The convolutional frontend consists of three 1D temporal convolution layers with channel sizes [128, 256, 256], kernel size 3, stride 2 in the first layer for temporal downsampling, and max pooling with kernel size 2. Each convolution layer is followed by BatchNorm and ReLU. The BiLSTM uses hidden size 256 per direction (512 output features total), with 2 stacked layers and dropout 0.1 between layers. Outputs are projected to character logits and trained with CTC loss. The model is optimized using Adam with learning rate 10^{-3} and cosine annealing scheduling.

Transformer. Transformer encoder with convolutional patching (`TransformerEncoderModule` + CTC head). The convolutional stem uses stride 4 to reduce temporal resolution before attention. The encoder uses $d_{\text{model}} = 256$, 6 encoder layers, 8 attention heads, feedforward dimension 1024 ($4 \times d_{\text{model}}$), and dropout 0.1. Learned positional embeddings are added to preserve temporal order, with sinusoidal fallback for long sequences. The encoder output is linearly projected to character logits and trained with CTC loss. Optimization uses Adam with learning rate 10^{-3} and cosine annealing scheduling.

Conv + Transformer (Adam). `TDSConvTransformerCTCModule`: $d_{\text{model}} = 384$, 8 heads, 4 layers, feedforward dim 1536, dropout 0.1, pre-LayerNorm. Adam optimizer, $\text{lr} = 10^{-3}$.

Conv + Transformer (AdamW + OneCycleLR). Identical encoder; AdamW $\text{lr} = 10^{-3}$, OneCycleLR with $\text{max_lr} = 10^{-3}$, 19050 total steps, 30% warmup, cosine annealing, $\text{div_factor} = 100$, $\text{final_div_factor} = 10^4$.

2.4 Channel Ablation

Three uniformly-spaced subsets are trained from scratch (in_features scaled accordingly, all other hyperparameters fixed): **4 ch** [0, 4, 8, 12], **6 ch** [0, 3, 6, 9, 12, 15], **8 ch** [0, 2, 4, 6, 8, 10, 12, 14].

2.5 Sampling rate.

We evaluated the impact of temporal resolution by varying the effective EMG sampling rate prior to spectrogram computation. Downsampling was implemented by adjusting the spectrogram hop length (with $n_{\text{fft}} = 64$ fixed), yielding effective rates of 62.5 Hz, 125 Hz (baseline), and 250 Hz. All other preprocessing steps, model architectures, and optimization settings were held constant.

2.6 Data volume.

To examine the effect of training data size, we trained models using 0.25, 0.50, 0.75, and 1.00 fractions of the available training sessions. Subsampling was performed at the session level using a fixed random seed to preserve label distribution. Validation and test splits, as well as all model and optimization hyperparameters, were kept unchanged across experiments.

3 Results

3.1 Architecture Comparison

Table 1: Architecture comparison. Test CER (%), lower is better.

Model	Val CER (%)	Test CER (%)	Rel. Δ vs. baseline
Baseline	32.48	28.33	—
GRU	17.77	18.56	+34.5%
BiLSTM	15.53	13.83	+51.2%
Transformer	18.19	20.89	+26.3%
Conv + Transformer (Adam)	14.02	30.21	−6.6%
Conv + Transformer (AdamW + OneCycleLR)	15.77	28.70	−1.3%

The BiLSTM achieves the lowest test CER of 13.83%, a 51.2% relative improvement over the baseline’s 28.33%. The GRU (4), despite sharing an identical TDS-Conv front-end (3) with the baseline, outperforms both Transformer variants and improves on the baseline by 34.5%. Both Conv+Transformer configurations fail to improve on the baseline, but the Adam variant performs worst overall at 30.21%, while AdamW+OneCycleLR (6; 7) marginally trails the baseline at 28.70%. Figure 1 shows the Conv+Transformer training dynamics: training loss collapses to near zero by epoch ~ 5 while validation CER remains volatile between 20–35%, confirming severe overfitting driven by a capacity-data mismatch rather than optimizer choice.

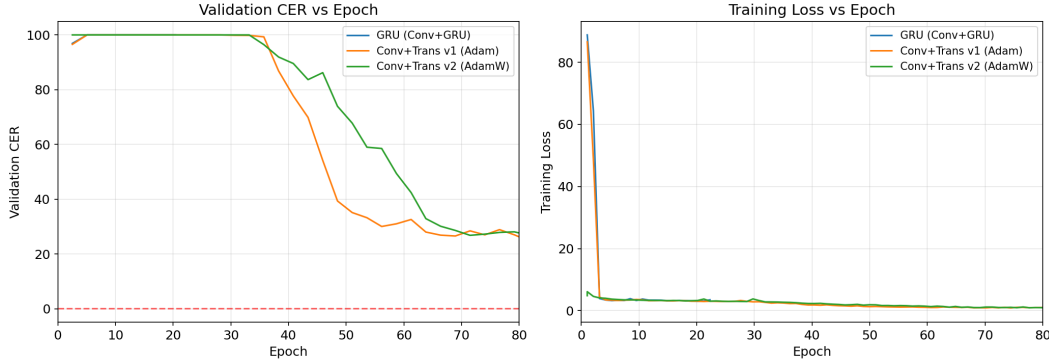


Figure 1: Validation CER (left) and training loss (right) for GRU, Conv+Transformer (Adam), and Conv+Transformer (AdamW+OneCycleLR). Training loss ≈ 0 by epoch ~ 5 for both Conv+Transformer variants; validation CER oscillates 20–35% throughout.

3.2 Channel Ablation

Table 2: Channel ablation: test CER (%) vs. electrode count (BiLSTM model).

Configuration	# Channels	Val CER (%)	Test CER (%)
stride4_4ch [0, 4, 8, 12]	4	27.89	25.87
stride3_6ch [0, 3, 6, 9, 12, 15]	6	20.16	19.69
stride2_8ch [0, 2, 4, 6, 8, 10, 12, 14]	8	18.43	16.51
<i>BiLSTM model</i>	16	15.53	13.83

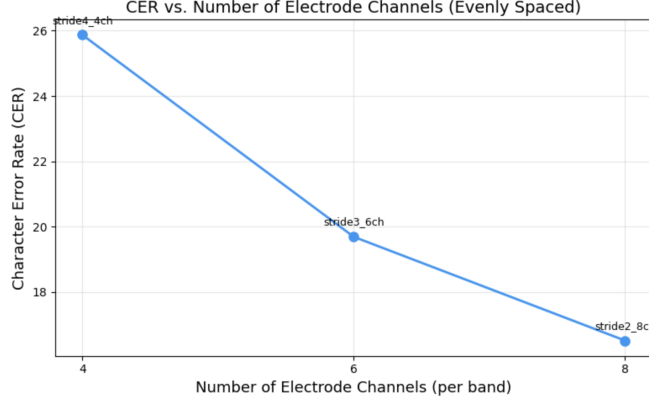


Figure 2: CER vs. electrode channel count. Gains diminish with density: the 4→6 step yields 6.18 points; 6→8 yields only 3.18 points. Even the 4-channel model outperforms the 16-channel baseline.

3.3 Sampling Rate

Model / Sampling Rate	Test CER (%)	Validation CER (%)
BiLSTM (125 Hz)	13.83	15.53
62.5 Hz	14.88	14.20
250 Hz	15.02	14.42

Table 3: Character Error Rate (CER) comparison across different EMG sampling rates using the BiLSTM model.

3.4 Data Volume

Train Size Fraction	Test CER (%)	Validation CER (%)
1.00	13.83	15.53
0.75	15.95	19.18
0.50	19.58	20.87
0.25	27.02	25.14

Table 4: Character Error Rate (CER) as a function of training data fraction.

4 Discussion

GRU outperforms Transformers: capacity-data mismatch. The baseline TDS-Conv achieves a test CER of 28.33% (Table 1). The standalone GRU improves this by 34.5%, while both Conv+Transformer (5) variants fail to beat the baseline. Adam increases test CER to 30.21% (−6.6% relative) and AdamW+OneCycleLR (6; 7) to 28.70% (−1.3% relative). The Conv+Transformer has ≈5.9M parameters versus the smaller capacity of the GRU, and with limited training data this excess capacity is spent on memorization rather than generalization, as confirmed by the near-zero training loss and volatile validation CER of 20–35% in Figure 1. The GRU’s smaller hypothesis class acts as a structural regularizer that the Transformer lacks without explicit weight decay or scheduling.

BiLSTM’s advantage over Baseline. The BiLSTM achieves the largest improvement, reducing test CER to 13.83%(Table 1), a 51.2% relative gain over the baseline’s 28.33%. The baseline uses only a TDS-Conv front-end followed by a linear MLP projection into the CTC decoder, with no dedicated sequence modeling component(3). The BiLSTM addresses this directly in two ways. First, bidirectional processing gives every output timestep access to both past and future context across the full 4-second window, whereas the baseline’s MLP projects each local TDS-Conv feature vector independently with no temporal memory. Second, the LSTM’s separate cell state c_t and hidden state h_t provide a two-pathway memory: the cell state acts as a long-term conveyor belt carrying information across keystroke boundaries with minimal gradient modification, while the hidden state provides the immediate output representation. The baseline has no equivalent mechanism to accumulate context across timesteps, so it cannot resolve ambiguities that depend on what came before or after a given window. For example, distinguishing pauses between repeated characters from a single held keystroke.

Adam vs. AdamW + OneCycleLR for the Transformer. After observing that the Conv+Transformer (5) trained with Adam overfit severely, the training loss collapsing to near zero by epoch 5 while validation CER remained high, AdamW+OneCycleLR (6; 7) was tested as a potential remedy. AdamW’s decoupled weight decay was intended to counteract unconstrained weight growth, and OneCycleLR’s cosine annealing schedule was expected to guide the optimizer toward flatter, more generalizable minima (9).

The results show a clear difference in convergence speed: Adam begins meaningfully reducing validation CER around epoch 35, while AdamW+OneCycleLR stays near 100% CER until epoch 45 due to OneCycleLR’s gradual warmup phase. Despite the slower start, both variants converge to a similar final CER of roughly 28–30%(Table 1) by epoch 80, and the training loss curves are nearly identical. This suggests that while AdamW+OneCycleLR (6) did not resolve the overfitting problem, the optimizer choice primarily affected convergence speed rather than final generalization performance, indicating the overfitting is more likely driven by a fundamental capacity-data mismatch than by the optimizer alone.

Channel ablation: diminishing returns and sparse-array feasibility. CER decreases monotonically with more electrodes as shown in Table 2 and Figure 2: 25.87% $\rightarrow$ 19.69% $\rightarrow$ 16.51% for 4, 6, 8 channels. The 4 $\rightarrow$ 6 step yields 6.18 points (23.9% relative), while 6 $\rightarrow$ 8 yields only 3.18 points (16.2% relative), consistent with diminishing marginal spatial information as inter-electrode spacing decreases. When neighboring electrodes are spaced 4 positions apart (4-channel config), the array likely misses intermediate recruitment signals from muscles between sampled sites. Denser coverage fills these gaps and provides the decoder with finer-grained spatial co-activation patterns that distinguish similar keystrokes (8). The asymptote visible in Figure 2 suggests that 8 channels approaches saturation of spatially useful information in this dataset. Most strikingly, the 4-channel model (25.87%) outperforms the 16-channel baseline (28.33%, Table 1), confirming that the baseline’s failure is driven by training factors rather than input richness. The 6-channel configuration is a practical sweet spot: 30.4% relative improvement over the 16-channel baseline at half the electrode count, and only 3.2 points above the 8-channel model.

Sampling rate. We evaluated the effect of EMG sampling rate on decoding performance by comparing 62.5 Hz, 125 Hz, and 250 Hz configurations(Table 3). Performance was best at 125 Hz, achieving a test CER of 13.83% and validation CER of 15.53%. At 62.5 Hz, the model achieves a worse test CER (14.88%) but better validation CER (14.20%), indicating overfitting. Similarly, at 250 Hz the model overfits with a worse test CER of 15.02% and better validation CER of 14.42%. Thus, sampling at 125 Hz is best, as it appears to prevent overfitting. This is consistent with the known sEMG power spectrum (8), which is concentrated in the 20–500 Hz band; sampling at 125 Hz is sufficient to capture most task-relevant temporal structure while avoiding unnecessary redundancy and noise amplification. Increasing the sampling rate beyond this range provides limited additional information for keystroke decoding, while downsampling too aggressively reduces the temporal resolution needed for accurate CTC alignment (2). Overall, 125 Hz provides a favorable trade-off between performance and computational efficiency.

Data volume. We analyzed the relationship between training data volume and decoding accuracy. As expected, reducing available training data increased CER monotonically, with performance

degrading substantially below 50% of the full dataset. While the model remains functional with as little as 25% of training sessions, the error rate nearly doubles compared to the full-data setting, indicating that sufficient coverage of typing variability is essential (Table 4). From a practical standpoint, these results suggest that moderate calibration data (e.g., 50–75% of the full dataset) may be sufficient for acceptable performance, but aggressive reductions significantly impair accuracy. This highlights a trade-off between user calibration burden and decoding quality.

Limitations. All results are from a single subject; architecture rankings and optimal channel counts may not transfer to other individuals. Ablations used fixed hyperparameters rather than re-tuning per condition. Each configuration used a single random seed, so differences below ~ 1 –2 CER points should be interpreted cautiously.

5 Conclusion

This work evaluated architectural and data-representation choices for single-subject sEMG keystroke decoding on the emg2qwerty benchmark across four axes: architecture search, channel ablation, sampling rate, and data volume.

Recurrent models outperform attention-based architectures in this small-data regime. The BiLSTM achieves the best test CER of 13.83%, a 51.2% relative improvement over the 28.33% baseline, while the GRU follows with a 34.5% gain. Both Conv+Transformer variants fail to surpass the baseline due to a fundamental capacity-data mismatch which neither AdamW nor OneCycleLR resolves, as optimizer choice affects convergence speed but not final generalization. The channel ablation shows diminishing marginal returns with electrode density, and notably the 4-channel sparse model still outperforms the 16-channel baseline, with the 6-channel configuration emerging as the practical sweet spot at half the electrode cost. Sampling at 125 Hz yields the best test CER by avoiding the overfitting seen at both 62.5 Hz and 250 Hz, and decoding quality degrades monotonically as training data is reduced, with the sharpest drop below 50% of the full dataset.

Taken together, practitioners should prioritize recurrent models and moderate electrode counts over large-capacity architectures or dense arrays, with 125 Hz sampling and at least 50–75% of available training data. As all results derive from a single subject with fixed hyperparameters and a single random seed, future work should validate these findings across multiple subjects and explore per-condition hyperparameter tuning and data augmentation to improve Transformer generalization at limited data scales.

A Hyperparameter Summary

Table 5: Hyperparameters for all architectures.

	GRU	BiLSTM	Transformer	Conv+Trans (Adam)	Conv+Trans (AdamW+OCL)
block_channels	[24,24,24,24]	—	—	[24,24,24,24]	[24,24,24,24]
kernel_width	32	3	3	32	32
mlp_features	[384]	[384]	[384]	[384]	[384]
Enc. hidden/ d_m	256	256	256	384	384
Enc. layers	4	2	6	4	4
nhead	—	—	8	8	8
FFN dim	—	—	1024	1536	1536
Dropout	0.0	0.1	0.1	0.1	0.1
Optimizer	Adam	Adam	Adam	Adam	AdamW
LR	10^{-3}	10^{-3}	10^{-3}	10^{-3}	10^{-3}
Scheduler	Cosine Annealing	Cosine Annealing	Cosine Annealing	Cosine Annealing	OneCycleLR
Epochs	80	80	80	80	80

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